

加入性别因素（进行one-hot编码），看看结果Acc值有没有变化

```
# 1 加入性别因素
```

```
import pandas as pd
import numpy as np # 一个熟悉的库, num+py
```

```
dataset = pd.read_csv('Social_Network_Ads.csv')
dataset.head()
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
accuracy = (cm[0,0]+cm[1,1])/(cm[0,0]+cm[0,1]+cm[1,0]+cm[1,1])
accuracy
```

0.95

=== 作业1: 加入性别因素 ===

混淆矩阵:

[[55 3]

[1 21]]

准确率: 95.00%

结论: 加入性别因素后准确率基本不变, 说明性别对购买行为影响不大

比较KNN与Logistic在这个数据案例上的结果差异，哪个算法对这个数据更有效

```
# ===== 作业2.1: KNN vs Logistic Regression =====

X = dataset.iloc[:, [2, 3]].values
y = dataset.iloc[:, 4].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)
sc = StandardScaler()
X_train_scaled = sc.fit_transform(X_train)
X_test_scaled = sc.transform(X_test)

# KNN
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)
knn_acc = accuracy_score(y_test, knn.predict(X_test_scaled))

# Logistic Regression
log_reg = LogisticRegression(random_state=0)
log_reg.fit(X_train_scaled, y_train)
log_acc = accuracy_score(y_test, log_reg.predict(X_test_scaled))

print("=== 作业2.1: KNN vs Logistic Regression ===")
print(f"KNN 准确率: {knn_acc * 100:.2f}%")
print(f"Logistic Regression 准确率: {log_acc * 100:.2f}%")
print(f"结论: {'KNN' if knn_acc > log_acc else 'Logistic Regression'} 表现更好\n")
```

=== 作业2.1: KNN vs Logistic Regression ===

KNN 准确率: 95.00%

Logistic Regression 准确率: 92.50%

结论: KNN 表现更好

训练集的比例，如何影响结果

```
# ===== 作业2.2: 训练集比例对结果的影响 =====

X = dataset.iloc[:, [2, 3]].values
y = dataset.iloc[:, 4].values

test_sizes = [0.1, 0.2, 0.3, 0.4, 0.5]
accuracies = []

for test_size in test_sizes:
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=test_size, random_state=0)
    sc = StandardScaler()
    X_train_scaled = sc.fit_transform(X_train)
    X_test_scaled = sc.transform(X_test)

    knn = KNeighborsClassifier(n_neighbors=5)
    knn.fit(X_train_scaled, y_train)
    acc = accuracy_score(y_test, knn.predict(X_test_scaled))
    accuracies.append(acc)

print("=== 作业2.2: 训练集比例影响 ===")
for i, ts in enumerate(test_sizes):
    print(f"测试集比例: {ts}, 准确率: {accuracies[i] * 100:.2f}%")

# 可视化
plt.figure(figsize=(10, 6))
plt.plot(test_sizes, accuracies, marker='o', linestyle='-', color='b')
plt.title('训练集比例对准确率的影响')
plt.xlabel('测试集比例')
plt.ylabel('准确率')
plt.grid(True)
plt.show()
```

=== 作业2.2: 训练集比例影响 ===

测试集比例: 0.1, 准确率: 92.50%

测试集比例: 0.2, 准确率: 95.00%

测试集比例: 0.3, 准确率: 91.67%

测试集比例: 0.4, 准确率: 89.38%

测试集比例: 0.5, 准确率: 88.50%

超参数K如何影响结果， 设置K=1、3、7、9

```
# ===== 作业2.3: 超参数K对结果的影响 =====

X = dataset.iloc[:, [2, 3]].values
y = dataset.iloc[:, 4].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)
sc = StandardScaler()
X_train_scaled = sc.fit_transform(X_train)
X_test_scaled = sc.transform(X_test)

k_values = [1, 3, 5, 7, 9]
accuracies = []

for k in k_values:
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_train_scaled, y_train)
    acc = accuracy_score(y_test, knn.predict(X_test_scaled))
    accuracies.append(acc)

print("=== 作业2.3: K值影响 ===")
for i, k in enumerate(k_values):
    print(f"K={k}, 准确率: {accuracies[i] * 100:.2f}%")

# 可视化
plt.figure(figsize=(10, 6))
plt.plot(k_values, accuracies, marker='o', linestyle='--', color='r')
plt.title('K值对准确率的影响')
plt.xlabel('K值')
plt.ylabel('准确率')
plt.xticks(k_values)
plt.grid(True)
plt.show()
```

=== 作业2.3: K值影响 ===

K=1, 准确率: 88.75%

K=3, 准确率: 95.00%

K=5, 准确率: 95.00%

K=7, 准确率: 95.00%

K=9, 准确率: 95.00%

对可视化图进行修改，同样是二维可视化的图，如果把性别变量考虑到可视化中（提示：可以用形状或者颜色区分）？

```
# ===== 作业3：包含性别因素的可视化 =====
```



```
dataset = pd.read_csv('Social_Network_Ads.csv')

# 按性别和购买状态分组
male_purchased = dataset[(dataset['Gender'] == 'Male') & (dataset['Purchased'] == 1)]
male_not = dataset[(dataset['Gender'] == 'Male') & (dataset['Purchased'] == 0)]
female_purchased = dataset[(dataset['Gender'] == 'Female') & (dataset['Purchased'] == 1)]
female_not = dataset[(dataset['Gender'] == 'Female') & (dataset['Purchased'] == 0)]

# 可视化
plt.figure(figsize=(12, 8))

# 男性 - 未购买（蓝色圆圈）
plt.scatter(male_not['Age'], male_not['EstimatedSalary'],
            c='blue', marker='o', label='男性-未购买', s=100, alpha=0.7)
# 男性 - 购买（蓝色星号）
plt.scatter(male_purchased['Age'], male_purchased['EstimatedSalary'],
            c='blue', marker='*', label='男性-购买', s=150, alpha=0.7)
# 女性 - 未购买（红色圆圈）
plt.scatter(female_not['Age'], female_not['EstimatedSalary'],
            c='red', marker='o', label='女性-未购买', s=100, alpha=0.7)
# 女性 - 购买（红色星号）
plt.scatter(female_purchased['Age'], female_purchased['EstimatedSalary'],
            c='red', marker='*', label='女性-购买', s=150, alpha=0.7)

plt.title('客户购买行为可视化（按性别区分）')
plt.xlabel('年龄')
plt.ylabel('预估收入')
plt.legend(loc='upper right')
plt.grid(True)
plt.show()
```